

# Time Series Analysis on Bearing's Accelerometry

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**Abstract**—In the world of robotics mechanical failure is prevalent. This paper showcases how time series analysis is able to detect anomalies on a bearing's vibration patterns from NASA's Bearing dataset and aid on the prevention of catastrophic failure.

## I. INTRODUCTION TO DATASET

The data was obtained from a mechanism meant to stress the bearing to failure point. Accelerometers and thermal sensors were placed on 4 bearings while being turned under a 6000 radial load. Time series analysis is applied for bearing 1 in one of the accelerometers to help identify patterns before the bearing fails. The data was sampled every 10 minutes at 20 kHz for 7 days as voltage readouts from the accelerometers. RMS was used to condense the 20 kHz [1].

## II. METHODS

### A. Cumulative Sum

Cumulative Sum (CUSUM) is widely used for anomaly detection. The model needs to learn the data before the anomaly happens, which can be achieved by calculating the mean  $\mu_0$  using the first 25% of the dataset:

$$S_i = \max(0, S_{i-1} + x_i - (\mu_0 + K)) \quad (1)$$

where  $S_i$  is the cumulative sum at time step  $i$ , with  $S_0 = 0$ ,  $x_i$  is the observed Root Mean Square (RMS) voltage of the accelerometer. The slack  $K$  is defined here:

$$K = 0.5\sigma_0 \quad (2)$$

where  $0.5\sigma_0$  standard deviation is also calculated from the first 25% of the dataset [2].

### B. Augmented Dickey-Fuller Test

Augmented Dickey-Fuller (ADF) test utilizes the best-fit slope from OLS divided by the standard error resulting in the T-statistic or ADF Score; this value is then evaluated under the Dickey-Fuller Distribution to conclude our final p-value. This test was performed at the end of each method:

$$t = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} \quad (3)$$

where  $\hat{\beta}_1$  is the estimated slope coefficient and  $SE(\hat{\beta}_1)$  is the standard error of that coefficient [3].

### C. Kwiatkowski-Phillips-Schmidt-Shin

In the process of Kwiatkowski-Phillips-Schmidt-Shin (KPSS) we first decompose the time series data as follows:

$$Y_t = \alpha + \delta t + r_t + \epsilon_t \quad (4)$$

where  $Y_t$  is the time series data,  $\alpha$  is the starting offset of the time series data,  $\delta t$  is the trend component,  $r_t$  is the stochastic drift component, and  $\epsilon_t$  is the random noise component. Stochastic drift is a prevalent pattern after a certain time in the dataset.

The residual component,  $e_i$ , is a combination of stochastic drift  $r_t$  and random noise  $\epsilon_t$ . We calculate the partial sum of the residuals as follows:

$$S_t = \sum_{i=1}^t e_i \quad (5)$$

the equation helps us conclude if the residuals are stable, otherwise indicating non-stationary.

To finalize, the KPSS test statistic (LM) is obtained by squaring and normalizing the partial sums against the background noise:

$$LM = \frac{\sum_{t=1}^T S_t^2}{T^2 \hat{\sigma}_\epsilon^2} \quad (6)$$

where  $T$  is the length of the time series data, and  $\hat{\sigma}_\epsilon^2$  is the squared variance. The data is more stationary if the result is closer zero [4].

### D. Transformations to Achieve Stationarity

Three main transformations can be applied to achieve stationarity after realizing non-stationary from ADF and KPSS:

- **Differencing:** The difference between the current and previous observations is calculated to mitigate the stochastic drift  $r_t$  and stabilize the mean of the series.
- **Logarithmic Transformation:** This transformation is applied to mitigate heteroscedasticity, or increasing variance, by compressing the scale of the vibration amplitudes.
- **Detrending:** The deterministic trend ( $\delta t$ ) is removed by subtracting the best-fit line obtained through Ordinary Least Squares (OLS) regression from the raw data, effectively centering the series [5].

### E. Friedman Test

Friedman test has the ability to highlight the most prevalent frequencies in a time series data set using the following equation:

$$Q = \frac{12n}{k(k+1)} \left[ \sum_{j=1}^k R_j^2 - \frac{k(k+1)^2}{4} \right] \quad (7)$$

where  $n$  is the Number of blocks (individual time snapshots),  $k$ : is the amount of seasonal groups (segments within a cycle), and  $R_j$  is the sum of ranks for the  $j$ -th group [6].

## III. RESULTS AND DISCUSSION

The structure of the following section will be comprised of the results in a figure or table following a discussion divided by sub headers.

### A. Moving Average

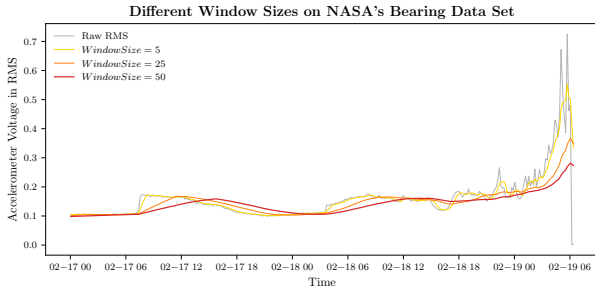


Fig. 1. Accelerometer Voltages in RMS plotted against different window sizes for moving average starting at Feb 17th

As the window size increases we can observe a clear loss of detail and resolution. Plotting with a window size of 50 is almost a flat line.

### B. CUSUM's Anomaly Detection

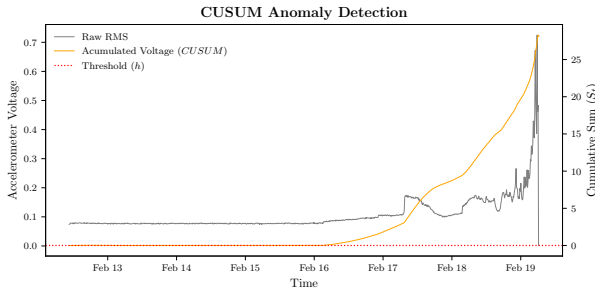


Fig. 2. CUSUM Anomaly Detection: Comparison on a dual axis graph of Raw RMS and the Accumulated Voltage Statistic ( $S_t$ ) with Anomaly Threshold ( $h$ ).

CUSUM was able to predict the anomaly 3 days in advance before the catastrophic failure. We can observe it wasn't some random noise because 3 days prior it remained flat because there wasn't anything to accumulate the sum.

### C. Evaluation of Transformations

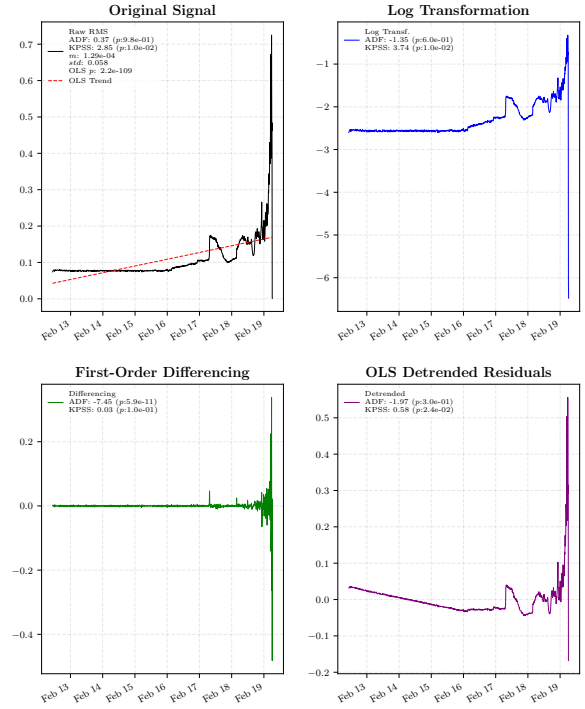


Fig. 3. 2x2 Grid of Time Series Transformations: (Top-Left) Original Signal with OLS Trend, (Top-Right) Log Transformation, (Bottom-Left) First-Order Differencing, and (Bottom-Right) OLS Detrended Residuals.

The most effective transformation to bring stationarity to the data was First-Order Differencing due to having the lowest ADF score and KPSS being closer to zero.

The ADF Score was calculated to be 0.3653, a positive value indicating that the signal deviation is significant and exhibits explosive growth characteristics. Consequently, the resulting P-value of 0.9802 fails to reject the null hypothesis, a result primarily driven by the exponential voltage oscillations observed towards the end of the dataset. (Ends here)

In OLS the slope is 0.000129 meaning that we have a trend in our data backed up with the standard deviation of 0.0582. Finally our P-value 2.2381e-109 means that we reject our null hypothesis of having no trend. Our data is not stationary.

### D. Statistical Tests for Stationarity

TABLE I  
STATIONARITY TEST RESULTS FOR RAW RMS SIGNAL

| Test | Statistic | p-value    | Crit. Value (5%) |
|------|-----------|------------|------------------|
| ADF  | 0.3653    | 9.8015e-01 | -2.8646          |
| KPSS | 0.5801    | 1.0000e-02 | 0.1460           |

Our differenced + RMS signal shows a significant improvement on the stationarity because ADF and KPSS is less than our critical value and the KPSS is closer to zero. The ADF p-value is lower, meaning its chance of not being stationary

TABLE II  
STATIONARITY TEST RESULTS FOR DIFFERENCED SIGNAL

| Test | Statistic | p-value    | Crit. Value (5%) |
|------|-----------|------------|------------------|
| ADF  | -7.4454   | 5.8567e-11 | -2.8646          |
| KPSS | 0.0273    | 1.0000e-01 | 0.4630           |

decrease. KPSS uses the opposite analogy due to the difference in the distributions, so the higher the p value means our values diverge from the estimated slope.

#### E. Hypothesis Testing for Seasonality

#### III.

TABLE III  
FRIEDMAN TEST RESULTS FOR SEASONALITY

| Metric               | Value      |
|----------------------|------------|
| Friedman Q Statistic | 39.2533    |
| p-value              | 1.8607e-02 |
| Degrees of Freedom   | 23         |

We got a high friedman score, meaning that our data is seasonal and with the p value being lower than 0.05 we can reject the null hypothesis.

#### IV. CONCLUSION

The highlight of this research was the RMS + CUSUM. The strategy was very effective detecting an anomaly 3 days before it happened and its a testament on how the most simplest solutions can have the best results.

#### V. REFERENCES

##### REFERENCES

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